

Free Energy Is a Signed, Zero-Training Detector of Acquisition Shift in Frozen Vision–Language Encoders: A Pre-Registered Falsification Study on Medical Image Corruptions

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Abstract

Frozen vision–language encoders such as CLIP enable zero-shot medical image classification in which class anchors are text embeddings and the softmax posterior over image–anchor distances is exactly a Gibbs distribution at temperature $T = 2\tau$, where τ is the temperature learned during contrastive pre-training. This identification makes two label-free observables available at no training cost: the Shannon entropy $\mathcal{H}$ of the posterior and the Helmholtz free energy F of the underlying energy model. Because a uniform translation of the anchor distances leaves $\mathcal{H}$ invariant but shifts F , we pre-registered a falsifiable prediction: under increasing acquisition shift, F separates source from corrupted images ($\text{AUROC} \geq 0.70$) while $\mathcal{H}$ stays near chance ($\text{AUROC} \leq 0.60$). We tested this prediction with a strictly zero-training protocol—prompts frozen before any target evaluation, temperature fixed at 2τ , every reported number traceable to a versioned run—on the official MedMNIST-C corruptions of BreastMNIST (ultrasound, seven corruptions) and PneumoniaMNIST (chest X-ray, four corruptions), with paired DeLong tests and stratified bootstrap confidence intervals. The strong form of the prediction is *refuted*: entropy is never near chance. What the data reveal instead is sharper. On structural corruptions (pixelation, JPEG, motion blur, speckle) F dominates $\mathcal{H}$ significantly (severity-5 AUROC up to 1.000 vs. 0.888; paired $\Delta\text{AUROC} +0.080$ to $+0.208$, all $p < 0.05$), whereas photometric corruptions (brightness, contrast, X-ray Gaussian noise) *decrease* F and drive the one-sided detector below or near chance (AUROC 0.127–0.594): free energy is a *signed* shift score. A two-sided variant $|F - \text{med}_{\text{source}}(F)|$, calibrated on source images only, restores above-chance detection on five of six sign-inverted cases (e.g., $0.127 \rightarrow 0.803$) while preserving the structural ones. Across prompt reformulations F is markedly more stable than $\mathcal{H}$ (severity-5 AUROC dispersion 0.03 vs. 0.08). All code, per-run metrics and raw scores are released; refuted criteria are reported as pre-registered.

1 Introduction

Zero-shot classification with frozen vision–language encoders (Radford et al., 2021) is an attractive substrate for medical imaging: class definitions are text prompts, no gradient touches the backbone, and the deployment surface is a single forward pass. Yet clinical images arrive from scanners, probes and acquisition protocols that drift over time and across sites, and a deployed zero-shot pipeline needs a label-free signal that the input distribution has moved (Rabanser et al., 2019). Such a signal is the gating quantity for every downstream mechanism studied in test-time adaptation (TTA) (Wang et al., 2021; Shu et al., 2022; Yoon et al., 2024; Niu et al., 2022, 2023): whether to adapt, when to stop, and which samples to refer to a human reader.

This paper studies the cheapest conceivable such signal. When image embeddings $\mathbf{z}$ are scored against K fixed text anchors $\boldsymbol{\mu}_k$, the CLIP posterior is a Gibbs (Boltzmann) distribution over squared distances at temperature $T = 2\tau$ (Section 2). Two observables then come for free: the prediction entropy $\mathcal{H}$ —the workhorse of TTA objectives and confidence baselines (Hendrycks and Gimpel, 2017; Wang et al., 2021)—and the Helmholtz free energy F , the log-partition function popularized for out-of-distribution detection on classifier logits (Liu et al., 2020; Grathwohl et al., 2020; LeCun et al., 2006) and later transposed to vision–language models (Ming et al., 2022).

Our starting point is a structural asymmetry between the two. If an acquisition shift moves images away from *all* anchors roughly uniformly — adding a constant c to every energy E_k — the posterior and hence $\mathcal{H}$ are unchanged, but F shifts by exactly c . This motivated a *pre-registered, falsifiable* prediction (criteria fixed before any target data were scored, Section 4): under increasing corruption severity, F detects the shift (AUROC ≥ 0.70) while $\mathcal{H}$ remains near chance (AUROC ≤ 0.60), with a margin of at least 0.05.

We tested this prediction under a deliberately austere protocol: no training, no adaptation, prompts frozen *a priori* and published verbatim, temperature fixed at the theoretical value 2τ , and every number in this paper generated by a versioned script and stored in a per-run metrics file. The testbed is MedMNIST-C (Di Salvo et al., 2024), the corruption benchmark built on MedMNIST v2 (Yang et al., 2023), on two modalities: breast ultrasound (BreastMNIST, derived from BUSI (Al-Dhabyani et al., 2020)) and chest X-ray (PneumoniaMNIST, derived from Kermany et al., 2018).

Findings. The strong form of the prediction is *refuted*: $\mathcal{H}$ is never near chance (severity-5 AUROC 0.462–0.888), because realistic corruptions do not translate energies uniformly. The refutation, however, uncovers a structure that we believe is more useful than the original conjecture:

1. **A structural/photometric dichotomy.** On corruptions that destroy spatial structure (pixelation, JPEG compression, motion blur, speckle noise), F dominates $\mathcal{H}$ significantly as a shift detector. On photometric corruptions (brightness, contrast, X-ray Gaussian noise), corrupted embeddings move *closer* to the text anchors, F decreases, and the one-sided detector drops below chance — to 0.127 in the worst case. Free energy is a *signed* score, not a magnitude.
2. **A source-calibrated two-sided rescue.** The statistic $|F - \text{med}_{\text{src}}(F)|$, using only the source median, restores above-chance detection on five of the six sign-inverted cases while leaving structural detection essentially intact.
3. **Prompt stability.** Across three prompt reformulations, the severity-5 detection AUROC of F varies by ~ 0.03 while that of $\mathcal{H}$ varies by ~ 0.08 : the partition function is more robust to the wording that defines the anchors than the posterior shape is.

We report the pre-registered verdict in full, including the criteria that failed, and release code, metrics and raw scores for every run.

2 Theoretical framework

2.1 Gibbs posterior over textual anchors

Let $\mathbf{z} \in \mathbb{R}^d$ be an ℓ_2 -normalized image embedding and $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K$ the ℓ_2 -normalized text embeddings of the class prompts. Define the configuration energy

$$E_k(\mathbf{z}) = \|\mathbf{z} - \boldsymbol{\mu}_k\|_2^2 = 2 - 2\langle \mathbf{z}, \boldsymbol{\mu}_k \rangle. \quad (1)$$

The Gibbs distribution at temperature $T > 0$ is $p_k(\mathbf{z}) = \exp(-E_k/T) / \sum_j \exp(-E_j/T)$. CLIP computes logits $\langle \mathbf{z}, \boldsymbol{\mu}_k \rangle / \tau$ with a learned temperature τ ; substituting (1) shows the two softmaxes coincide exactly when

$$T = 2\tau. \quad (2)$$

No temperature is tuned in this work: T is read off the frozen model (here $T \approx 0.0200$).

2.2 Observables and the thermodynamic identity

We track the entropy, free energy, mean energy and a normalized order parameter (“crystallinity”):

$$\mathcal{H}(\mathbf{z}) = -\sum_k p_k \log p_k, \quad F(\mathbf{z}) = -T \log \sum_k e^{-E_k/T}, \quad \langle E \rangle(\mathbf{z}) = \sum_k p_k E_k, \quad (3)$$

$$\chi(\mathbf{z}) = 1 - \mathcal{H}(\mathbf{z}) / \log K, \quad \text{with the identity} \quad F = \langle E \rangle - T \mathcal{H} \quad (4)$$

used as a software invariant (unit-tested to numerical precision). F is, up to the distance parameterization, the negative energy score of Liu et al. (2020) evaluated on distance logits.

2.3 The pre-registered prediction

Proposition 1 (Translation asymmetry). *For any constant c , the map $E_k \mapsto E_k + c$ leaves p and therefore $\mathcal{H}$ and χ unchanged, while $F \mapsto F + c$.*

If acquisition shift pushed embeddings away from all anchors while roughly preserving relative class separations, F would detect it and $\mathcal{H}$ would not. Before scoring any corrupted image we therefore froze the following decision criteria, evaluated at maximum severity with the source cohort as negatives:

- C1. $\text{AUROC}(F) \geq 0.70$;
- C2. $\text{AUROC}(\mathcal{H}) \leq 0.60$;
- C3. $\text{AUROC}(F) - \text{AUROC}(\mathcal{H}) \geq 0.05$.

The prediction is *confirmed* only if all three hold. These thresholds are decision rules, not results; a cleanly executed refutation is reported as such.

3 Related work

Energy-based OOD detection. Liu et al. (2020) showed that the logit log-partition function outperforms maximum softmax probability (Hendrycks and Gimpel, 2017) for OOD detection; Grathwohl et al. (2020) read classifiers as energy-based models. Ming et al. (2022) transposed confidence-based OOD scores to CLIP. Our contribution is orthogonal to score engineering: a pre-registered, severity-resolved comparison of F against $\mathcal{H}$ on *medical acquisition-style* corruptions, exposing the signed behavior of F that global OOD benchmarks average away.

Dataset-shift detection. Rabanser et al. (2019) compared two-sample tests on learned representations. We restrict attention to the two statistics that are free in a deployed zero-shot pipeline, and quantify their per-sample discriminative power (AUROC) rather than cohort-level test power.

Test-time adaptation. Entropy minimization is the dominant TTA objective (Wang et al., 2021; Niu et al., 2022, 2023), extended to prompt tuning in vision-language models (Shu et al., 2022) with calibration corrections (Yoon et al., 2024). Our results bear on the *gating* of such methods: a signal that moves under shift while entropy barely moves (or moves for the wrong reason) is the natural trigger and stopping observable. This paper establishes the zero-training baseline for that program; the adaptation stage itself is out of scope here.

Benchmarks. MedMNIST-C (Di Salvo et al., 2024) provides modality-specific corruptions at five severities for MedMNIST v2 (Yang et al., 2023); we use its official corruption operators verbatim.

4 Experimental protocol

4.1 Substrate

Encoder: CLIP ViT-B/16 (OpenAI weights, QuickGELU variant) via `open_clip` (Ilharco et al., 2021), entirely frozen, run in float32 on CPU (PyTorch 2.4.1, Python 3.11, Windows). Temperature fixed at $T = 2\tau \approx 0.0200$ per (2). Anchors are the ℓ_2 -normalized means of the per-class prompt-template embeddings (prompt ensembling), renormalized after averaging.

4.2 Cohorts and corruptions

BreastMNIST (Yang et al., 2023; Al-Dhabyani et al., 2020): breast ultrasound, binary malignant vs. normal/benign, test split $n = 156$ at 224×224 (MedMNIST+). All seven official MedMNIST-C corruptions for this dataset: `speckle_noise`, `motion_blur`, `brightness_up`, `brightness_down`, `contrast_down`, `jpeg_compression`, `pixelate`, at severities 0 (clean source) through 5. **PneumoniaMNIST** (Kermany et al., 2018): chest X-ray, binary normal vs. pneumonia, test split $n = 624$; corruptions `gaussian_noise`, `gaussian_blur`, `brightness_down`, `contrast_down`.

Zero-shot diagnostic performance on clean data is deliberately reported and modest — BreastMNIST balanced accuracy 0.517 (near chance; accuracy 0.712, AUROC 0.636), PneumoniaMNIST accuracy 0.620 — the shift detection question is independent of diagnostic quality, and no claim of clinical utility of the zero-shot classifier is made.

4.3 Prompts, frozen a priori

The prompt sets were fixed before any corrupted image was scored and are reproduced verbatim. BreastMNIST classes {“a malignant breast tumor”, “a normal or benign breast tissue”} with templates {“a breast ultrasound image of {}.”, “an ultrasound scan showing {}.”, “grayscale sonography of {}.”}; PneumoniaMNIST classes {“a normal chest”, “a chest with pneumonia”} with templates {“a chest x-ray image of {}.”, “a radiograph showing {}.”}. Prompt sensitivity is measured by re-running the full protocol with each single template (Section 5.4).

4.4 Metrics and statistics

For each severity $s \geq 1$ we compute the AUROC of F (and of $\mathcal{H}$) as a per-sample discriminator between the clean cohort ($s = 0$, negatives) and the corrupted cohort (positives); higher score should indicate the corrupted domain. Because both scores are computed on the same samples, we compare them with the paired DeLong test (DeLong et al., 1988) and a stratified paired bootstrap (Efron and Tibshirani, 1994) with 10^4 resamples for 95% confidence intervals on $\Delta\text{AUROC} = \text{AUROC}(F) - \text{AUROC}(\mathcal{H})$. The two-sided variant replaces a sample’s score x by

Table 1: Severity-5 shift detection (source vs. corrupted), canonical runs (prompt ensemble, seed 0, severities 0–5). $\Delta = \text{AUROC}(F) - \text{AUROC}(\mathcal{H})$ with stratified paired bootstrap 95% CI and paired DeLong p -value. $|F|_2 = \text{two-sided variant } |F - \text{med}_{\text{src}}(F)|$. Structural corruptions above the rule, photometric below.

Dataset	Corruption	AUROC(F)	AUROC($\mathcal{H}$)	Δ [95% CI]	p	$ F _2$
Breast	pixelate	1.000	0.888	+0.112 [+0.078, +0.150]	1.9×10^{-9}	1.000
Breast	jpeg_compression	0.920	0.711	+0.208 [+0.143, +0.274]	7.7×10^{-10}	0.899
Breast	motion_blur	0.885	0.760	+0.125 [+0.060, +0.190]	1.7×10^{-4}	0.845
Breast	speckle_noise	0.814	0.734	+0.080 [+0.006, +0.154]	0.032	0.742
Pneumonia	gaussian_blur	0.970	0.798	+0.172 [+0.148, +0.198]	3.9×10^{-41}	0.962
Breast	brightness_up	0.594	0.677	−0.083 [−0.176, +0.012]	0.083	0.447
Breast	brightness_down	0.359	0.563	−0.204 [−0.297, −0.112]	1.3×10^{-5}	0.606
Breast	contrast_down	0.299	0.498	−0.199 [−0.287, −0.112]	7.4×10^{-6}	0.630
Pneumonia	gaussian_noise	0.127	0.462	−0.335 [−0.368, −0.300]	1.7×10^{-86}	0.803
Pneumonia	brightness_down	0.200	0.644	−0.444 [−0.481, −0.405]	1.6×10^{-117}	0.593
Pneumonia	contrast_down	0.209	0.698	−0.489 [−0.526, −0.452]	2.9×10^{-148}	0.584

$|x - \text{med}_{\text{src}}|$ where med_{src} is the median of the *source* scores; no target information enters the calibration.

4.5 Leakage controls and traceability

(i) Prompts frozen before target evaluation; (ii) $T = 2\tau$, never tuned; (iii) no threshold is fit on target data (the two-sided center uses source only); (iv) every number below carries a run identifier; each run stores its configuration, git commit, code hash, environment, per-severity metrics and raw per-sample scores. A caveat we cannot close: CLIP’s web-scale pre-training corpus is not auditable, so memorization of the public source images cannot be excluded; corruption operators, however, are applied locally and cannot have been seen.

5 Results

5.1 Pre-registered verdict: the strong prediction is refuted

Table 1 reports severity-5 detection for all canonical runs. Criterion C2 — entropy near chance — fails on *every* corruption (minimum severity-5 AUROC($\mathcal{H}$) = 0.462, maximum 0.888). The conjunction $C1 \wedge C2 \wedge C3$ is therefore refuted in all configurations, as recorded in each run’s verdict block (`prediction_confirmed: false`). Realistic corruptions do not translate the energy landscape uniformly: they also reshape relative distances, which entropy sees.

5.2 Free energy is a signed detector

The refutation has a clean geometry. On structural corruptions the corrupted embeddings move *away* from all text anchors: mean F rises monotonically with severity (BreastMNIST speckle: $1.288 \rightarrow 1.351$; pixelate: $1.288 \rightarrow 1.479$) and F dominates $\mathcal{H}$ significantly (Table 1, upper block; all DeLong $p < 0.05$, CIs excluding zero). On photometric corruptions the embeddings move *closer* to the anchors — mean F *decreases* (contrast_down: $1.288 \rightarrow 1.266$; PneumoniaMNIST gaussian_noise: $1.264 \rightarrow 1.234$) — and the one-sided AUROC falls below 0.5, to 0.127 in the extreme case. A plausible reading is that brightness/contrast compression and moderate X-ray noise push images toward the generic, high-density region of CLIP’s image manifold, reducing the image–text modality gap; we do not claim a mechanism beyond the observed sign.

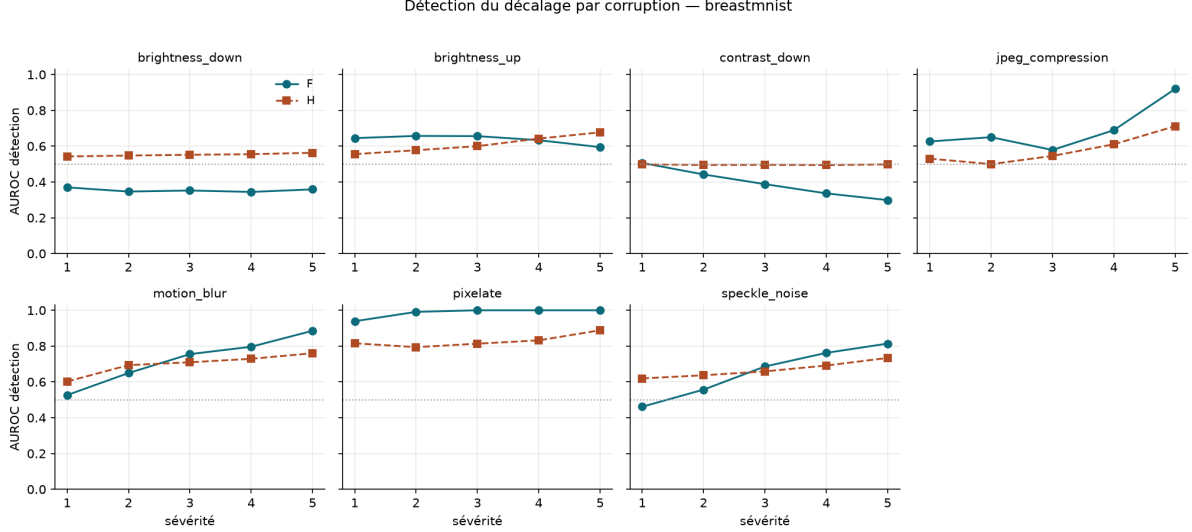


Figure 1: Severity-resolved detection AUROC of F (solid) and $\mathcal{H}$ (dashed) for the seven official BreastMNIST-C corruptions. Dotted line: chance. Note the below-chance trajectories of F on `brightness_down` and `contrast_down`.

Two practical consequences follow. First, using raw F (or its negation, as in standard energy-score OOD pipelines) as a shift alarm silently fails on exactly the photometric drifts most typical of scanner and probe settings. Second, the failure is directional, hence recoverable.

5.3 A source-calibrated two-sided score rescues the inversions

The statistic $|F - \text{med}_{\text{src}}(F)|$ uses only source images. At severity 5 it restores above-chance detection on five of the six sign-inverted cases — PneumoniaMNIST `gaussian_noise` $0.127 \rightarrow 0.803$, BreastMNIST `contrast_down` $0.299 \rightarrow 0.630$, `brightness_down` $0.359 \rightarrow 0.606$, PneumoniaMNIST `brightness_down` $0.200 \rightarrow 0.593$ and `contrast_down` $0.209 \rightarrow 0.584$ — while leaving structural cases essentially unchanged (`pixelate` 1.000, `jpeg` 0.899, `motion_blur` 0.845, `gaussian_blur` 0.962). The recovery is strong where the downward drift is large (X-ray Gaussian noise) and modest where the drift is shallow relative to source dispersion (X-ray photometric shifts, ≈ 0.59 , where $\mathcal{H}$ itself reaches 0.64–0.70). The residual failure, `brightness_up` (0.447), is a genuinely hard case: its F trajectory is non-monotonic (up at low severity, down at high), so no function of F alone can order it; its entropy signal (0.677) and the X-ray photometric cases together suggest a bivariate $(F, \mathcal{H})$ score as the natural next step.

5.4 Prompt sensitivity: F is more stable than $\mathcal{H}$

The anchors are literally defined by prompt wording, so prompt sensitivity is the substrate-specific confound. Re-running the full protocol with each single template (BreastMNIST, severity 5):

Corruption	AUROC(F)				AUROC($\mathcal{H}$)			
	t0	t1	t2	ens.	t0	t1	t2	ens.
<code>speckle_noise</code>	0.778	0.801	0.854	0.814	0.777	0.641	0.597	0.734
<code>brightness_down</code>	0.351	0.419	0.307	0.359	0.580	0.464	0.609	0.563

The dispersion of F across templates (s.d. 0.032 on `speckle`, 0.046 on `brightness_down`) is roughly half that of $\mathcal{H}$ (0.076 and 0.064): the partition function aggregates the anchor geometry

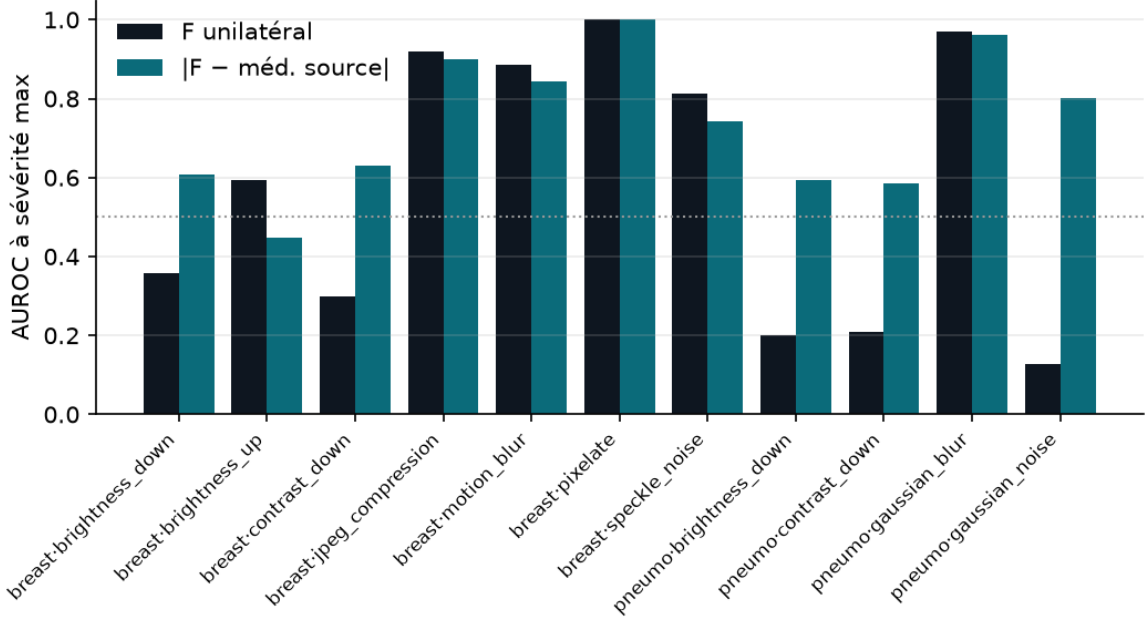


Figure 2: One-sided F vs. two-sided $|F - \text{med}_{\text{src}}(F)|$ detection AUROC at maximum severity, all canonical runs.

and is less exposed to a particular wording than the posterior shape. The sign inversion on brightness_down persists under *every* template — it is not a prompt artifact.

5.5 Robustness to corruption sampling

Corruption operators are stochastic. Across three seeds (BreastMNIST, speckle_noise, severity 5): $\text{AUROC}(F) = 0.814/0.819/0.827$, $\text{AUROC}(\mathcal{H}) = 0.734/0.749/0.738$; the paired severity-5 ΔAUROC ranges $+0.069$ to $+0.088$ with overlapping CIs. Sampling noise is an order of magnitude smaller than the effects reported above.

5.6 Second modality

The chest X-ray cohort replicates the dichotomy on an independent modality, anatomy and corruption implementation, with $4\times$ larger test sets ($n = 624$). Gaussian blur (structural) behaves like the ultrasound structural block: F rises monotonically ($1.264 \rightarrow 1.320$) and dominates $\mathcal{H}$ (0.970 vs. 0.798 , $\Delta = +0.172$, $p = 3.9 \times 10^{-41}$). All three photometric corruptions invert the sign of F — Gaussian noise, brightness and contrast compression each move X-ray embeddings *closer* to the text anchors (F AUROC 0.127 – 0.209), with effect sizes far beyond sampling noise ($|\Delta| \geq 0.335$, $p \leq 10^{-86}$). On this modality the inversions are deeper than in ultrasound, and $\mathcal{H}$ is the better detector for the two photometric-compression cases (0.644 and 0.698): neither observable subsumes the other, which is precisely the argument for reporting both and for the bivariate score sketched above.

6 Discussion and limitations

What the refutation teaches. The pre-registered conjecture assumed acquisition shift acts as a quasi-uniform translation of the energy landscape. The data say otherwise: realistic corruptions move embeddings along the modality-gap axis in *either* direction and simultaneously reshape relative distances. The corrected picture — F as a signed coordinate of drift along the

anchor-distance axis, $\mathcal{H}$ as a shape statistic — is more useful than the original conjecture, and directly actionable via the two-sided score.

Implications for test-time adaptation. Entropy-minimizing TTA (Wang et al., 2021; Shu et al., 2022) adapts wherever entropy is high. Our results exhibit shifts where entropy barely moves while F moves strongly (JPEG: $\mathcal{H}$ AUROC 0.711 vs. F 0.920) and shifts where entropy rises while F falls. A gating rule based on $|F - \text{med}_{\text{src}}|$ is the zero-cost candidate this study puts forward for the adaptation-triggering and early-stopping stages of our broader research program on frozen vision–language encoders in ultrasound.

Limitations. (i) Test cohorts are small ($n = 156$ ultrasound images); CIs are reported accordingly. (ii) Single backbone (CLIP ViT-B/16, OpenAI weights); a biomedical encoder (Zhang et al., 2023) may shift the geometry and is the designated robustness check. (iii) MedMNIST-C corruptions are synthetic surrogates for acquisition shift; the multi-center ultrasound transfer (BUSI→UDIAT→BUS-UCLM) is the pre-registered next cohort. (iv) The two-sided median is computed on the full source cohort (in-sample); with $n = 156$ and one degree of freedom the optimism is negligible, but a held-out calibration split is the clean design. (v) Images are 224×224 upsamplings of MedMNIST sources. (vi) CLIP pre-training contamination cannot be audited (Section 4).

7 Conclusion

Under a zero-training, leakage-controlled, pre-registered protocol on official medical corruption benchmarks, the Helmholtz free energy of a frozen CLIP posterior is a strong but *signed* detector of acquisition shift: significantly better than entropy on structural corruptions, below chance on photometric ones, and repaired in five of six failure cases by a two-sided source-calibrated variant that requires nothing but source images. The strong form of our pre-registered prediction — entropy at chance — is refuted and reported as such. Every number in this paper is traceable to a versioned run; code, metrics and raw scores are released with the paper.

Reproducibility. All experiments: single CPU, no training; ~ 3 minutes per corruption sweep (156 images $\times$ 6 severities). Code, per-run `metrics.json`, raw scores and analysis scripts are available in the project repository.

References

- Walid Al-Dhabyani, Mohammed Gomaa, Hussien Khaled, and Aly Fahmy. Dataset of breast ultrasound images. *Data in Brief*, 28:104863, 2020.
- Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3):837–845, 1988.
- Francesco Di Salvo, Sebastian Doerrich, and Christian Ledig. Medmnist-c: Comprehensive benchmark and improved classifier robustness by simulating realistic image corruptions. *arXiv preprint arXiv:2406.17536*, 2024.
- Bradley Efron and Robert J. Tibshirani. An introduction to the bootstrap. 1994.
- Will Grathwohl, Kuan-Chieh Wang, Jörn-Henrik Jacobsen, David Duvenaud, Mohammad Norouzi, and Kevin Swersky. Your classifier is secretly an energy based model and you should treat it like one. In *International Conference on Learning Representations (ICLR)*, 2020.

- Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017.
- Gabriel Ilharco, Mitchell Wortsman, Ross Wightman, Cade Gordon, Nicholas Carlini, Rohan Taori, Achal Dave, Vaishal Shankar, Hongseok Namkoong, John Miller, Hannaneh Hajishirzi, Ali Farhadi, and Ludwig Schmidt. Openclip, 2021. URL https://github.com/mlfoundations/open_clip.
- Daniel S. Kermany, Michael Goldbaum, Wenjia Cai, Carolina C. S. Valentim, Huiying Liang, Sally L. Baxter, Alex McKeown, Ge Yang, Xiaokang Wu, Fangbing Yan, et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5):1122–1131, 2018.
- Yann LeCun, Sumit Chopra, Raia Hadsell, Marc’Aurelio Ranzato, and Fu Jie Huang. A tutorial on energy-based learning. In *Predicting Structured Data*. MIT Press, 2006.
- Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 21464–21475, 2020.
- Yifei Ming, Ziyang Cai, Jiuxiang Gu, Yiyao Sun, Wei Li, and Yixuan Li. Delving into out-of-distribution detection with vision-language representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 35087–35102, 2022.
- Shuaicheng Niu, Jiayang Wu, Yifan Zhang, Yaofo Chen, Shijian Zheng, Peilin Zhao, and Minghui Tan. Efficient test-time model adaptation without forgetting. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pages 16888–16905, 2022.
- Shuaicheng Niu, Jiayang Wu, Yifan Zhang, Zhiquan Wen, Yaofo Chen, Peilin Zhao, and Minghui Tan. Towards stable test-time adaptation in dynamic wild world. In *International Conference on Learning Representations (ICLR)*, 2023.
- Stephan Rabanser, Stephan Günnemann, and Zachary C. Lipton. Failing loudly: An empirical study of methods for detecting dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021.
- Manli Shu, Weili Nie, De-An Huang, Zhiding Yu, Tom Goldstein, Anima Anandkumar, and Chaowei Xiao. Test-time prompt tuning for zero-shot generalization in vision-language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 14274–14289, 2022.
- Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations (ICLR)*, 2021.
- Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. Medmnist v2 – a large-scale lightweight benchmark for 2d and 3d biomedical image classification. *Scientific Data*, 10(1):41, 2023.

Hee Suk Yoon, Eunseop Yoon, Joshua Tian Jin Tee, Mark Hasegawa-Johnson, Yingzhen Li, and Chang D. Yoo. C-tpt: Calibrated test-time prompt tuning for vision-language models via text feature dispersion. In *International Conference on Learning Representations (ICLR)*, 2024.

Sheng Zhang, Yanbo Xu, Naoto Usuyama, Hanwen Xu, Jaspreet Bagga, Robert Tinn, Sam Preston, Rajesh Rao, Mu Wei, Naveen Valluri, Cliff Wong, Andrea Tupini, Yu Wang, Matt Mazzola, Swadheen Shukla, Lars Liden, Jianfeng Gao, Matthew P. Lungren, Tristan Naumann, and Hoifung Poon. Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint arXiv:2303.00915*, 2023.